

Neural Net Report

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Question 5: Learning With Restarts

1. testPenData:

- Max accuracy: 0.9073756432246999
- Average accuracy: 0.9012006861063465
- Standard deviation: 0.007332301818881595

2. testCarData:

- Max accuracy: 0.985
- Average accuracy: 0.981
- Standard deviation: 0.0037416573867739447

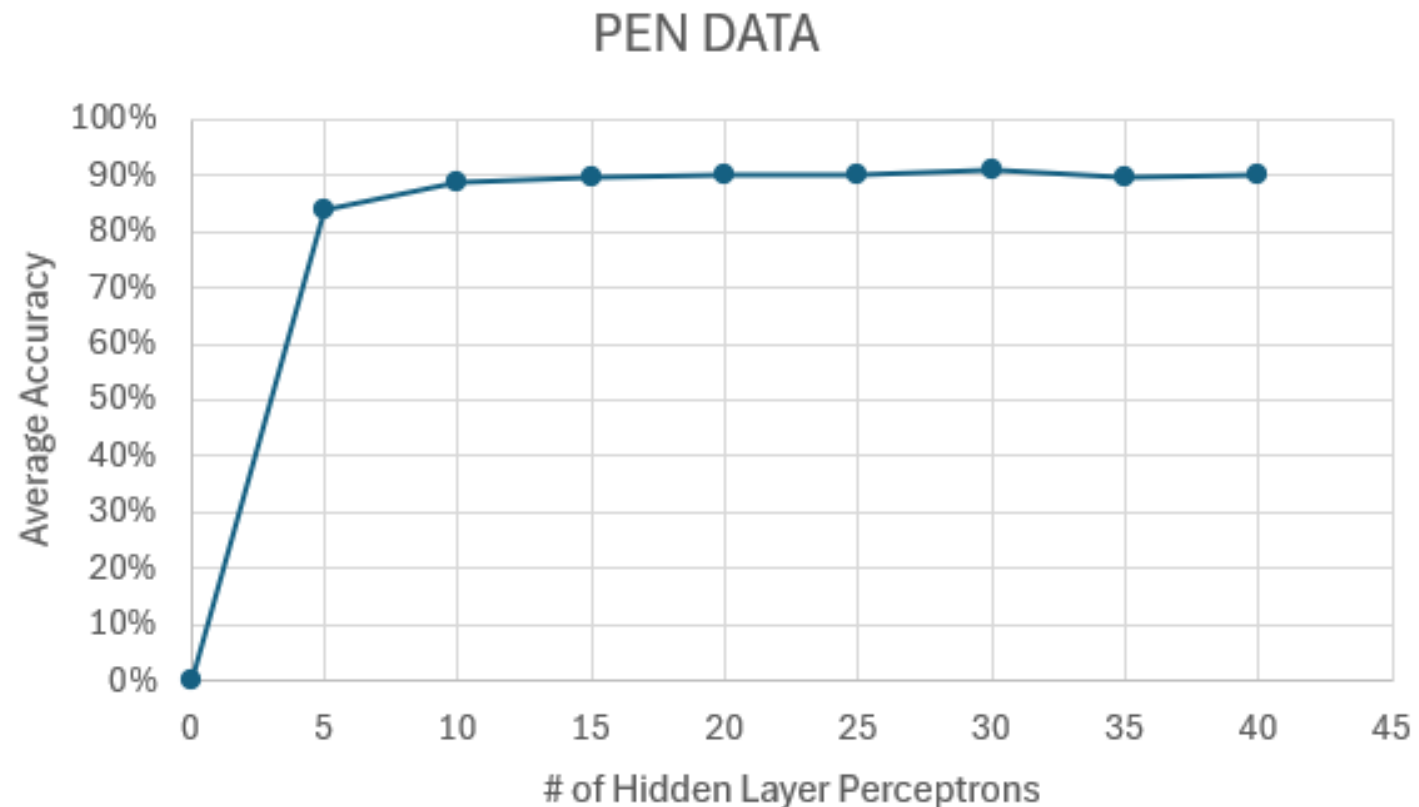
Question 6: Varying the Hidden Layers

Statistic table for **testPenData** – report the max, average, and standard deviation at various amount of perceptrons.

	Number of Perceptrons at the Hidden Layer								
	0	5	10	15	20	25	30	35	40
Max Accuracy	0.000000	0.856204	0.892224	0.903373	0.908805	0.909377	0.940537	0.908519	0.907947
Avg Accuracy	0.000000	0.840652	0.887421	0.897999	0.902802	0.902459	0.912007	0.898571	0.900915
Standard Deviation	0.000000	0.014712	0.006233	0.005918	0.005152	0.008022	0.014345	0.008221	0.005211

Question 6: Varying the Hidden Layers

Create a learning curve for **testPenData** where the number of hidden layer perceptrons is the independent variable and the average accuracy is the dependent variable.



Question 6: Varying the Hidden Layers

For **testPenData**, discuss any notable trends you saw related to increasing the size of the hidden layers in your neural net.

Answer: TestPenData started off at 0% average accuracy with no hidden layers and grew dramatically from 0 → 5 hidden layers. However, as the number of hidden layers increases past 10, there is not a significant improvement in the average accuracy. The graph demonstrates the turbulence in the data, as illustrated by the very slight rising and falling of the average accuracy.

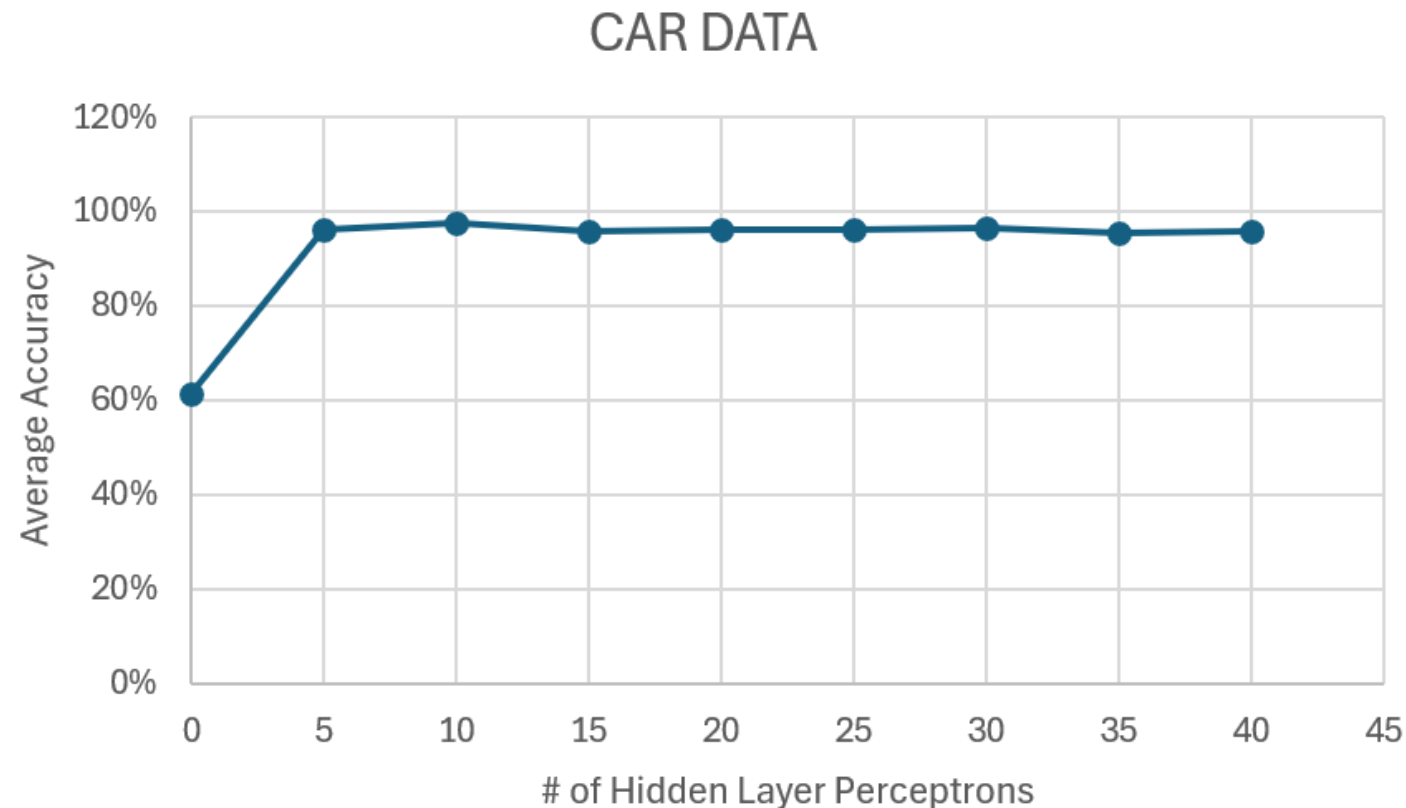
Question 6: Varying the Hidden Layers

Statistic table for **testCarData** – report the max, average, and standard deviation at various amount of perceptrons.

	Number of Perceptrons at the Hidden Layer								
	0	5	10	15	20	25	30	35	40
Max Accuracy	0.615	0.990000	0.985000	0.970000	0.965000	0.970000	0.990000	0.965000	0.965000
Avg Accuracy	0.615	0.962000	0.976000	0.960000	0.961000	0.963000	0.966000	0.957000	0.958000
Standard Deviation	0.000000	0.016310	0.008602	0.007071	0.005831	0.006000	0.018547	0.005099	0.005099

Question 6: Varying the Hidden Layers

Create a learning curve for **testCarData** where the number of hidden layer perceptrons is the independent variable and the average accuracy is the dependent variable.



Question 6: Varying the Hidden Layers

For **testCarData**, discuss any notable trends you saw related to increasing the size of the hidden layers in your neural net.

Answer: Similarly to testPenData, we find the average increased the greatest with from 0 $\rightarrow$ 5 permutations. However, no hidden layer did not result in the average accuracy being 0, which was the primary difference between the two datum. Furthermore, we still see a turbulence effect, however, the range of such is less than that of PenData and the average accuracy seems to be higher compared to PenData.

Question 7 (extra credit): Learning XOR

Report the max accuracy, average accuracy, and standard deviation of the neural net that you have trained with 1) no hidden layer, and 2) a hidden layer with various amount of perceptrons (at least 3 different amounts)

	No Hidden Layer	Hidden Layer		
		5 perceptrons	10 perceptrons	15 perceptrons
Max Accuracy	0.5	0.5	0.75	0.75
Avg Accuracy	0.3	0.3	0.5	0.6
Standard Deviation	0.244948	0.1	0.15811	0.2

Question 7 (extra credit): Learning XOR

Report the behavior of the trained neural net **without a hidden layer**.

Answer: No hidden layer results in a low average accuracy of 0.3. This is to be expected since a neural network without a hidden layer can only model linearly separable function (only linear relationships). Overall, the performance is poor as a result of it not fitting the non-linear nature of XOR.

Question 7 (extra credit): Learning XOR

Report the behavior of the trained neural net **with a hidden layer**. Are the results what you expected? Explain your observation.

Answer: The behavior is better and follows what was expected, with increasing perceptions and number of neutrons in the hidden layer corresponding to a greater accuracy. The greater the neutrons, the more connections are able to be made. I did not, however, expect to need 48 neutrons/perceptions to reach 100% accuracy. When learning rate is increased, though, the number of necessary neutrons falls considerably.

Question 8 (extra credit): Novel Dataset

List the name and the source of the dataset that you've chosen.

- Name: _____
- Source (e.g., URLs): _____
- Briefly describe the dataset: _____

Question 8 (extra credit): Run Stats

- Max accuracy: _____
- Average accuracy: _____
- Standard deviation: _____

Question 8 (extra credit): Novel Dataset

Describe how to run the code that you've set up to train the selected dataset.

Answer: